Project Proposal: Telco Customer Churn Prediction

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1. Dataset description

Source: Kaggle, "Telco Customer Churn" (IBM sample data), https://www.kaggle.com/datasets/blastchar/telco-customer-churn. The raw file has 7,043 rows and 21 columns, well above the 2,000-row requirement. Each row is one telecom customer. Features include demographics (gender, senior citizen, partner, dependents), account information (tenure, contract type, paperless billing, payment method, monthly and total charges) and subscribed services (phone, multiple lines, internet type, online security, backup, device protection, tech support, streaming). The target column is **Churn** (Yes/No), indicating whether the customer left within the last month.

2. Why this dataset suits the project

- Mix of numeric and categorical features, so encoding and one-hot encoding are meaningful.
- Hidden missing values: TotalCharges is stored as text with 11 blank entries for brand-new customers.
- Imbalanced target (73.5% No / 26.5% Yes), which makes class balancing necessary.
- TotalCharges is right-skewed, which justifies normalization; numeric columns have different ranges, which justifies scaling.
- Clear business value and a well-defined supervised learning problem for the finals.

3. What I will accomplish (pre-processing, all in Python / Jupyter)

- Split the data 70% train / 30% test in code. EDA and all fitted steps use training data only; the test set stays untouched.
- EDA with Matplotlib: class distribution, histograms, boxplots, churn rate by category, correlation heatmap.
- Fill nulls (median for skewed numeric values, mode for categorical), using training statistics.
- Feature engineering: tenure groups, average monthly spend, charge change, number of services, support bundle flag, autopay flag.
- Encoding: binary label encoding, ordinal encoding for Contract and tenure group, one-hot encoding for nominal columns.
- Normalization (log transform of skewed features) and scaling (StandardScaler fit on train only).
- Class balancing with SMOTE on the training set only. Before/after plots and sample rows are shown at each step.
- No manual edits to the data at any point; the clean datasets are saved from code.

4. Machine learning plan for the finals

Binary classification to predict churn so a company can target retention offers. I plan to compare Logistic Regression, Random Forest, Gradient Boosting and KNN/SVM, tuned with cross-validation on the training data. Because the test set keeps its natural class ratio, I will evaluate with precision, recall, F1 for the churn class, ROC-AUC and a confusion matrix rather than accuracy alone, and inspect feature importance to explain which factors drive churn.